

JIANGTAO KONG

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🎓 EDUCATION

College of William & Mary, Williamsburg, VA 09/2021 – Present

MS/PhD Computer Science

Beijing University of Technology, Beijing, China 09/2015 – 07/2019

Bachelor Electronic Information Engineering

📖 RESEARCH AREAS

- Continual Learning, Multimodal Models, World Models, World-Model-Based Online RL

📄 SELECTED PUBLICATIONS

- **ECA: Efficient Continual Alignment for Open-Ended Image-to-Text Generation.**
Jiangtao Kong, et al. (*ICML 2026*)
- **WestWorld: A Knowledge-Encoded Scalable Trajectory World Model for Diverse Robotic Systems.**
Yuchen Wang*, Jiangtao Kong*, et al. (*ICML 2026 Spotlight*)
- **ODESteer: A Unified ODE-Based Steering Framework for LLM Alignment.**
Hongjue Zhao*, Haosen Sun*, Jiangtao Kong, et al. (*ICLR 2026*)
- **YoooP: You Only Optimize One Prototype per Class for Non-Exemplar Incremental Learning.**
Jiangtao Kong, et al. (*TMLR 2025*)
- **Hybrid Memory Replay: Blending Real and Distilled Data for Class Incremental Learning.**
Jiangtao Kong, et al. (*arXiv 2024*)

*Equal contribution.

👥 RESEARCH EXPERIENCE

World-model-based online RL for Long-Horizon Planning 2025 – Present

Current Research World-Model-Based Online RL

- Developing a world-model-based online RL framework for long-horizon planning, extending step-level latent dynamics and policies to trajectory-level world models and action-chunk policies.
- Designing trajectory-level policy optimization where action chunks are proposed, evaluated through latent rollouts, and improved using rollout rewards and terminal value estimates under online model updates.
- Studying online adaptation challenges in trajectory-level world-model RL, including compounding rollout error, planner-induced distribution shift, and evolving policy targets.

JPMorgan Chase & Co., New York, NY 06/2024 – 08/2024; 09/2024 – 12/2024

Summer Research Associate / Part-time Researcher Leader: Shaohan Hu

- Proposed **ECA**, an efficient continual alignment framework for open-ended image-to-text generation, accepted to *ICML 2026*.
- Designed parameter-efficient adaptation modules for pretrained vision-language models, including query mixing, Fisher-guided expansion, and dictionary replay.
- Built incremental multimodal benchmarks for captioning and VQA-style generation, evaluating continual adaptation under evolving visual and semantic distributions.

Institute of Automation, Chinese Academy of Sciences, Beijing, China 10/2020 – 06/2021

Research Intern Leader: Jun Wan

- Conducted research on person/vehicle re-identification and face detection, including model adaptation and deployment for computer vision systems.